Лабораторная работа 4

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Цель работы

Получение навыков правильной работы с репозиториями git.

Задание

Выполнить работу для тестового репозитория. Преобразовать рабочий репозиторий в репозиторий с git-flow и conventional commits.

Выполнение работы

Установка git-flow

```
Enabling a Copr repository. Please note that this repository is not part
of the main distribution, and quality may vary.

The Fedora Project does not exercise any power over the contents of
this repository beyond the rules outlined in the Copr FAQ at
<https://docs.pagure.org/copr.copr/user_documentation.html#what-i-can-build-in-copr>,
and packages are not held to any quality or security level.

Please do not file bug reports about these packages in Fedora
Bugzilla. In case of problems, contact the owner of this repository.
Is this ok [y/N]: y
mgmizinov@localhost-live:~$ sudo dnf install gitflow
Updating and loading repositories:
 Fedora 43 openh264 (From C 100% |   1.5 KiB/s |   5.8 KiB |  00m04s
 Copr repo for gitflow owne 100% |   0.0   B/s |   1.5 KiB |  00m00s
 Fedora 43 - x86_64 - Updat  69% |   6.0 MiB/s |   9.2 MiB |  00m00 Copr repo for gitflow own 100% | 392.0   B/s |   2.4 KiB |  0
 Fedora 43 - x86_64 - Upda 100% |   1.6 MiB/s |  12.8 MiB |  00m08s Fedora 43 - x86_64        100% |   2.4 MiB/s |  35.4 MiB |  0
Repositories loaded.
Package             Arch   Version             Repository      Size
Installing:
 gitflow            x86_64 1.12.3-1.fc34       copr:copr. 261.7 KiB

Transaction Summary:
 Installing:         1 package

Total size of inbound packages is 57 KiB. Need to download 57 KiB.
After this operation, 262 KiB extra will be used (install 262 KiB, remove 0 B).
Is this ok [y/N]: y
[1/1] gitflow-0:1.12.3-1.fc34.x86_64     100% |  57.4 KiB |  00m00s
--------------------------------------------------------------------------------
[1/1] Total                              100% |  57.4 KiB |  00m00s
[1/2] https://download.copr.fedorainfrac ???% |   0.0   B |  00m00s
[1/2] https://download.copr.fedorainfrac ???% |   0.0   B |  00m00s
[1/2] https://download.copr.fedorainfrac 100% | 998.0   B |  00m00s
```

1 рисунок

Установка Node.js

```
mgmizinov@localhost-live:~$ sudo dnf install nodejs
Updating and loading repositories:
Repositories loaded.
Package             Arch   Version               Reposito      Size
Installing:
 nodejs             x86_64 1:22.22.0-2.fc43      updates  161.6 KiB
Installing dependencies:
 nodejs-libs        x86_64 1:22.22.0-2.fc43      updates   78.1 MiB
Installing weak dependencies:
 nodejs-docs        noarch 1:22.22.0-2.fc43      updates   97.5 MiB
 nodejs-full-i18n   x86_64 1:22.22.0-2.fc43      updates   30.4 MiB
 nodejs-npm         x86_64 1:10.9.4-1.22.22.0.2. updates    9.2 MiB

Transaction Summary:
 Installing:         5 packages

Total size of inbound packages is 40 MiB. Need to download 40 MiB.
After this operation, 215 MiB extra will be used (install 215 MiB, remove 0 B).
Is this ok [y/N]: y
[1/5] nodejs-1:22.22.0-2.fc43.x86_64     100% |  51.4 KiB |  00m02s
[2/5] nodejs-docs-1:22.22.0-2.fc43.noarc 100% |   9.1 MiB |  00m02s
[3/5] nodejs-full-i18n-1:22.22.0-2.fc43. 100% |   8.5 MiB |  00m03s
[4/5] nodejs-npm-1:10.9.4-1.22.22.0.2.fc 100% |   2.3 MiB |  00m05s
[5/5] nodejs-libs-1:22.22.0-2.fc43.x86_6 100% |  19.6 MiB |  00m30s
--------------------------------------------------------------------------------
[5/5] Total                              100% |  39.7 MiB |  00m31s
Running transaction
Importing OpenPGP key 0x31645531:
 UserID     : "Fedora (43) <fedora-43-primary@fedoraproject.org>"
 Fingerprint: C6E7F081CF80E13146676E88829B606631645531
 From       : file:///etc/pki/rpm-gpg/RPM-GPG-KEY-fedora-43-x86_64
Is this ok [y/N]: y
The key was successfully imported.
[1/7] Verify package files               100% |   5.0   B |  00m00s
[2/7] Prepare transaction                100% |   5.0   B |  00m00s
[3/7] Installing nodejs-libs-1:22.22.0-2 100% |  78.1 MiB |  00m00s
[4/7] Installing nodejs-1:22.22.0-2.fc43 100% | 163.7 KiB |  00m00s
[5/7] Installing nodejs-npm-1:10.9.4-1.2 100% |   9.6 MiB |  00m00s
[6/7] Installing nodejs-full-i18n-1:22.2 100% |  30.4 MiB |  00m00s
[7/7] Installing nodejs-docs-1:22.22.0-2 100% |  97.6 MiB |  00m07s
Complete!
mgmizinov@localhost-live:~$ sudo dnf install pnpm
Updating and loading repositories:
```

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Настройка Node.js

```
mgmizinov@localhost-live:~$ pnpm setup
Appended new lines to /home/mgmizinov/.bashrc

Next configuration changes were made:
export PNPM_HOME="/home/mgmizinov/.local/share/pnpm"
case ":$PATH:" in
  *":$PNPM_HOME:"*) ;;
  *) export PATH="$PNPM_HOME:$PATH" ;;
esac

To start using pnpm, run:
source /home/mgmizinov/.bashrc
mgmizinov@localhost-live:~$ source ~/.bashrc
mgmizinov@localhost-live:~$
```

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Общепринятые коммиты commitizen и standard-changelog

```
mgmizinov@localhost-live:~$ source ~/.bashrc
mgmizinov@localhost-live:~$ pnpm add -g commitizen


    Update available! 10.27.0 → 10.30.3.
    Changelog: https://pnpm.io/v/10.30.3
      To update, run: pnpm self-update


 WARN  2 deprecated subdependencies found: glob@7.2.3, inflight@1.0.6
Packages: +150
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
Progress: resolved 150, reused 0, downloaded 150, added 150, done

/home/mgmizinov/.local/share/pnpm/global/5:
+ commitizen 4.3.1

Done in 5.6s using pnpm v10.27.0
mgmizinov@localhost-live:~$ pnpm add -g standard-changelog
 WARN  2 deprecated subdependencies found: glob@7.2.3, inflight@1.0.6
Packages: +30
++++++++++++++++++++++++++++++
Progress: resolved 180, reused 150, downloaded 30, added 30, done

/home/mgmizinov/.local/share/pnpm/global/5:
+ standard-changelog 7.0.1

Done in 3.1s using pnpm v10.27.0
mgmizinov@localhost-live:~$
```

4 рисунок

Практический сценарий использования git Создание репозитория git

1) Подключение репозитория к github

```
mgmizinov@localhost-live:~/git-extended$ git commit -m "first commit"
On branch main

Initial commit

nothing to commit (create/copy files and use "git add" to track)
mgmizinov@localhost-live:~/git-extended$ git remote add origin git@github.com:<username>/git-extended.git
bash: username: No such file or directory
mgmizinov@localhost-live:~/git-extended$ git remote add origin git@github.com:MihailMizinov/git-extended.git
error: remote origin already exists.
mgmizinov@localhost-live:~/git-extended$ git push -u origin master
```

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2) Конфигурация общепринятых коммитов

```
mgmizinov@localhost-live:~/git-extended$ git add .
mgmizinov@localhost-live:~/git-extended$ git cz
cz-cli@4.3.1, cz-conventional-changelog@3.3.0

? Select the type of change that you're committing: feat:     A new
 feature
? What is the scope of this change (e.g. component or file name):
(press enter to skip)
? Write a short, imperative tense description of the change (max 94
 chars):
 (8) new file
? Provide a longer description of the change: (press enter to skip)

? Are there any breaking changes? Yes
? A BREAKING CHANGE commit requires a body. Please enter a longer
description of the commit itself:
 -
? Describe the breaking changes:
 new file
? Does this change affect any open issues? No
[main 4f12377] feat: new file
 1 file changed, 14 insertions(+)
 create mode 100644 package.json
mgmizinov@localhost-live:~/git-extended$ git push
Username for 'https://github.com': MihailMizinov
```

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3) Конфигурация git-flow

```
mgmizinov@localhost-live:~/git-extended$ standard-changelog --first-release
mgmizinov@localhost-live:~/git-extended$ git add CHANGELOG.md
git commit -am 'chore(site): add changelog'
[release/1.0.0 c0890a0] chore(site): add changelog
 1 file changed, 11 insertions(+)
 create mode 100644 CHANGELOG.md
mgmizinov@localhost-live:~/git-extended$ git flow release finish 1.0.0
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
Merge made by the 'ort' strategy.
 CHANGELOG.md | 11 +++++++++++
 1 file changed, 11 insertions(+)
 create mode 100644 CHANGELOG.md
Already on 'main'
Your branch is ahead of 'origin/main' by 2 commits.
  (use "git push" to publish your local commits)
fatal: no tag message?
fatal: Tagging failed. Please run finish again to retry.
mgmizinov@localhost-live:~/git-extended$ git push --all
git push --tags
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 578 bytes | 578.00 KiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To github.com:MihailMizinov/git-extended.git
   4f12377..8f1b314  main -> main
 * [new branch]      release/1.0.0 -> release/1.0.0
Everything up-to-date
mgmizinov@localhost-live:~/git-extended$ gh release create v1.0.0 -F CHANGELOG.md
bash: gh: command not found...
mgmizinov@localhost-live:~/git-extended$ sudo dnf install gh
Updating and loading repositories:
Repositories loaded.
Package                                         Arch        Version                          Repository                          Size
Installing:
 gh                                             x86_64      2.87.0-2.fc43                    updates                           37.6 MiB

Transaction Summary:
 Installing:        1 package

Total size of inbound packages is 11 MiB. Need to download 11 MiB.
After this operation, 38 MiB extra will be used (install 38 MiB, remove 0 B).
Is this ok [y/N]: y
[1/1] gh-0:2.87.0-2.fc43.x86_64                                                   100% |   3.6 MiB/s |  11.0 MiB |  00m03s
--------------------------------------------------------------------------------------------------------------------------------
[1/1] Total                                                                       100% |   2.5 MiB/s |  11.0 MiB |  00m04s
Running transaction
[1/3] Verify package files                                                        100% |   8.0   B/s |   1.0   B |  00m00s
[2/3] Prepare transaction                                                         100% |   2.0   B/s |   1.0   B |  00m00s
[3/3] Installing gh-0:2.87.0-2.fc43.x86_64                                        100% |  58.0 MiB/s |  37.7 MiB |  00m01s
Complete!
mgmizinov@localhost-live:~/git-extended$ gh release create v1.0.0 -F CHANGELOG.md
```

7 рисунок

Работа с репозиторием git

1) Разработка новой функциональности

```
mgmizinov@localhost-live:~/git-extended$ git flow feature start feature_branch
Switched to a new branch 'feature/feature_branch'

Summary of actions:
- A new branch 'feature/feature_branch' was created, based on 'develop'
- You are now on branch 'feature/feature_branch'

Now, start committing on your feature. When done, use:

     git flow feature finish feature_branch

mgmizinov@localhost-live:~/git-extended$ git flow feature finish feature_branch
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
Already up to date.
Deleted branch feature/feature_branch (was 4f12377).

Summary of actions:
- The feature branch 'feature/feature_branch' was merged into 'develop'
- Feature branch 'feature/feature_branch' has been locally deleted
- You are now on branch 'develop'

mgmizinov@localhost-live:~/git-extended$
```

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2) Создание релиза git-flow

```
mgmizinov@localhost-live:~/git-extended$ standard-changelog
mgmizinov@localhost-live:~/git-extended$ git add CHANGELOG.md
git commit -am 'chore(site): update changelog'
[release/1.2.3 17aa813] chore(site): update changelog
 1 file changed, 1 insertion(+)
mgmizinov@localhost-live:~/git-extended$ git flow release finish 1.2.3
Branches 'develop' and 'origin/develop' have diverged.
And local branch 'develop' is ahead of 'origin/develop'.
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
Merge made by the 'ort' strategy.
 CHANGELOG.md | 1 +
 1 file changed, 1 insertion(+)
Already on 'main'
Your branch is ahead of 'origin/main' by 4 commits.
  (use "git push" to publish your local commits)
fatal: no tag message?
Fatal: Tagging failed. Please run finish again to retry.
mgmizinov@localhost-live:~/git-extended$ git flow release finish 1.2.3
Branches 'main' and 'origin/main' have diverged.
And local branch 'main' is ahead of 'origin/main'.
Branches 'develop' and 'origin/develop' have diverged.
And local branch 'develop' is ahead of 'origin/develop'.
Already on 'main'
Your branch is ahead of 'origin/main' by 4 commits.
  (use "git push" to publish your local commits)
Switched to branch 'develop'
Your branch is ahead of 'origin/develop' by 3 commits.
  (use "git push" to publish your local commits)
Merge made by the 'ort' strategy.
 CHANGELOG.md | 1 +
 1 file changed, 1 insertion(+)
Deleted branch release/1.2.3 (was 17aa813).

Summary of actions:
- Release branch 'release/1.2.3' has been merged into 'main'
- The release was tagged '1.2.3'
- Release tag '1.2.3' has been back-merged into 'develop'
- Release branch 'release/1.2.3' has been locally deleted
- You are now on branch 'develop'

mgmizinov@localhost-live:~/git-extended$ git push --all
git push --tags
Enumerating objects: 8, done.
Counting objects: 100% (8/8), done.
Delta compression using up to 2 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 867 bytes | 867.00 KiB/s, done.
Total 6 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To github.com:MihailMizinov/git-extended.git
   a6cc6b6..b87f892  develop -> develop
   8f1b314..b7e390f  main -> main
Enumerating objects: 2, done.
Counting objects: 100% (2/2), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 298 bytes | 298.00 KiB/s, done.
Total 2 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:MihailMizinov/git-extended.git
 * [new tag]         1.0.0 -> 1.0.0
 * [new tag]         1.2.3 -> 1.2.3
mgmizinov@localhost-live:~/git-extended$ gh release create v1.2.3 -F CHANGELOG.md
https://github.com/MihailMizinov/git-extended/releases/tag/v1.2.3
```

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Вывод

Полученил необходимые навыки правильной работы с репозиториями git.